Iformer: Irregular Spatial–Spectral Transformer for Hyperspectral Image Classification

Muhammad Ahmad and Manuel Mazzara

***Abstract*—The hyperspectral image (HSI) classification (HSIC) remains challenging due to severe class imbalance dominated by background pixels, thin and edge-adjacent structures with anisotropic context, nonstationary spectral responses across materials, and illumination variability in large HSI cubes. The existing convolutional and transformer-based methods often rely on stationary kernels and uniform spectral sampling, which limits their ability to model irregular spatial–spectral dependencies. This work introduces Iformer, a band-aware irregular spatial–spectral transformer that incorporates coordinate-constrained spectral modeling and geometry-aware spatial propagation. The proposed framework is built upon three core components. First, the irregular spectral aggregation (ISA) models the spectral axis using coordinate-aware anchors with content-gated attention, enabling adaptation to spectral nonstationarity while reducing spectral tokens ($S' \rightarrow R_{\text{ref}} \ll S'$) to control computational complexity. Second, irregular spatial propagation (ISA_{spa}) introduces a coordinate-driven anisotropic kernel to capture directional spatial context, improving discrimination of fine and structurally complex regions. Third, irregular multihead self-attention (iMHSA) enables heterogeneous subspace allocation across attention heads, allowing the model to adapt its representational capacity to varying spectral–spatial complexities. A unified spatial–spectral embedding supports these components via a 3-D patchifier with channel reweighting and a calibrated CLS+GAP readout to improve prediction stability under class imbalance. Extensive experiments on four widely used benchmarks demonstrate that Iformer consistently achieves superior accuracy.**

***Index Terms*—Anisotropic context, boundary fidelity, hyperspectral image classification (HSIC), irregularities, multihead self-attention (MHSA).**

I. Introduction

HYPERSPECTRAL imaging (HSI) extends conventional imaging by capturing hundreds of contiguous spectral bands per pixel, transforming each scene into a spatially organized set of high-dimensional reflectance spectra [1]. This rich spectral representation enables fine-grained material discrimination and semantic mapping [2], supporting applications in land-cover analysis, vegetation and soil assessment, urban analytics, hydrology, and environmental monitoring [3].

Received 14 February 2026; revised 12 April 2026 and 25 May 2026; accepted 2 July 2026. Date of publication 6 July 2026; date of current version 21 July 2026. This work was supported by the Ministry of Economic Development of the Russian Federation under Agreement 139-10-2025-034 dd. 19.06.2025 and Agreement IGK 000000C313925P4D0002. *(Corresponding author: Muhammad Ahmad.)*

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Digital Object Identifier 10.1109/TGRS.2026.3710374

The central challenge in HSI classification (HSIC) is to jointly model spatial context on the image grid and structured dependencies along the spectral axis [4], while operating under limited labeled data and constrained computational resources [5].

Despite substantial progress, HSIC remains challenging due to several intrinsic factors. First, datasets are severely imbalanced, with a dominant background class and limited samples for minority categories [6], [7], [8]. Second, spatial structures are often anisotropic, consisting of thin, fragmented, and edge-adjacent objects whose contextual support is direction-dependent rather than isotropic [9], [10], [11]. Third, spectral responses are nonstationary across materials and spatial locations, with irregular band spacing and intraclass variability due to illumination and mixing effects [12], [13]. Finally, the high dimensionality of HSI cubes imposes significant computational and memory constraints, particularly for attention-based models with quadratic complexity [14], [15]. These challenges necessitate models that can adapt to irregular spatial–spectral structure while maintaining efficiency and robustness.

A. Conventional and Domain-Specific Methods

Early HSIC approaches relied on pixelwise spectral classifiers, such as nearest neighbor, logistic regression, support vector machines, and shallow neural networks [16], often combined with dimensionality reduction techniques, including PCA [17], LDA [18], and manifold embeddings [19]. While effective in exploiting spectral information, these methods underutilize spatial context and rely on stationary assumptions that are poorly suited to complex scene structures. Extensions using morphological profiles and graphical models improve spatial consistency but depend on hand-crafted priors and may oversmooth fine structures and minority classes [20], [21], [22], [23]. Moreover, these methods do not explicitly encode spectral ordering and spacing, limiting their ability to model structured spectral relationships [24].

B. Convolutional Neural Networks (CNNs)

The CNN-based methods introduce learnable spatial–spectral feature extraction through 1-D, 2-D, and 3-D convolutions [25]. While 3-D CNNs better integrate spatial and spectral information, they incur high computational cost and remain constrained by stationary, isotropic kernels [26], [27]. Such kernels are mismatched to anisotropic spatial structures and irregular spectral variations, leading to loss of fine details and degraded performance on minority classes [28], [29]. Although attention-CNNs partially address these limitations, they typically operate as channel reweighting and local modulation rather than fully adaptive, coordinate-aware modeling

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of spatial–spectral interactions [20]. Furthermore, many CNN-based pipelines lack principled handling of normalization, boundary effects, and class imbalance, which can reduce robustness and calibration [30].

C. Transformer-Based Methods

Transformers model global dependencies by representing HSIs as token sequences or grids [31]. Spectral transformers treat bands as tokens [32], while spatial and hybrid variants operate on patchified representations [33], [34], [35]. Although attention mechanisms enable long-range modeling, the existing approaches often rely on uniform tokenization and generic positional encodings, ignoring physically meaningful spectral ordering and irregular band spacing [36]. In addition, the standard multihead self-attention (MHSA) enforces uniform head dimensions, limiting representational flexibility and increasing computational cost when token counts are large [37], [38]. Spectral token counts often scale linearly with B, leading to quadratic complexity in attention [39], [40]. Moreover, issues, such as boundary handling, per-sample normalization, class imbalance, and calibration, are inconsistently addressed [41], [42], [43].

A unified view of prior work reveals persistent gaps: spectral-only models underuse spatial structure [44]; CNNs rely on stationary kernels that struggle with anisotropic patterns [45], [46]; transformers treat spectral positions as exchangeable tokens and lack coordinate-aware modeling [47]; and many pipelines overlook normalization, boundary handling, and imbalance-aware training, leading to unstable optimization and poor calibration [48], [49], [50], [51].

To address these limitations, this work proposes Iformer, a band-aware irregular spatial–spectral transformer that introduces coordinate-aware modeling and irregular aggregation mechanisms. The key idea is to replace uniform aggregation and stationary attention with irregular, coordinate-constrained operations that better align with the underlying geometry of HSI data. The main contributions of this work are summarized as follows.

1) *Irregular spectral modeling via coordinate-aware aggregation:* We introduce irregular spectral aggregation (ISA), which performs content-gated aggregation over a small set of learned spectral anchors using normalized band coordinates. This formulation captures spectral nonstationarity and reduces token complexity from S' to $R_{\text{ref}} \ll S'$.
2) *Geometry-driven spatial propagation for anisotropic context:* We propose irregular spatial propagation (ISA_{spa}), a coordinate-based kernel that models directional spatial dependencies, enabling improved representation of thin and edge-adjacent structures. Unlike isotropic convolutions and content-only attention, ISA_{spa} explicitly models directional affinities on the spatial grid, preserving geometric continuity and improving robustness to anisotropic context.
3) *Irregular multihead attention with heterogeneous capacity allocation:* We develop irregular MHSA (iMHSA), which relaxes the uniform-head constraint by allowing heterogeneous query/key dimensions across heads, enabling specialization across complementary spatial–spectral subspaces. Combined with Pre-Norm, LayerScale, and DropPath, iMHSA provides more stable and expressive long-range modeling.
4) *Unified spatial–spectral framework with calibrated learning:* We integrate the above components into a unified transformer architecture with efficient embedding (i.e., embedding based on a single Conv3−D operator over (u, v, b), followed by squeeze-and-excitation (SE) to adaptively emphasize informative channels). This design jointly encodes local spectral and spatial context into a uniform embedding grid, avoiding the decoupling inherent in separable 2-D/1-D pipelines and mitigating noisy-band dominance common in shallow or hybrid CNN stacks, stable optimization, and imbalance-aware prediction, achieving improved accuracy, robustness, and computational efficiency across diverse HSI benchmarks.

II. Methodology

Let the HSI be denoted by $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ with a corresponding label map $\mathbf{Y} \in \{0, 1, \ldots, C\}^{H\times W}$, where 0 indicates background and $1, \ldots, C$ denotes foreground classes.[1] To construct training samples, we extract fixed-size spatial patches of size $W_S \times W_S \times B$ centered at labeled pixels using dense sliding windows over a reflect-padded image. Reflection padding is employed to avoid boundary bias and to ensure that all patches, including those near image borders, share a consistent receptive field. Depending on the experimental setting, background pixels are excluded and retained via a binary foreground mask, yielding a set of valid patch centers and corresponding class labels. Each extracted patch is stored with a singleton channel dimension to align with 3-D convolutions, and labels are mapped to one-hot encodings with a consistent effective class count across training, validation, and test splits. To stabilize optimization under varying illumination and contrast, we apply per-sample, per-band Z-score standardization over spatial dimensions, ensuring that each patch is normalized independently without introducing train–test leakage. This preprocessing pipeline yields a compact patch tensor $\mathbf{P} \in \mathbb{R}^{N\times W_S\times W_S\times B\times 1}$[2] and associated labels, which serve as the input to the proposed spatial–spectral modeling framework.

A. 3-D Patchifier Convolution and Channel Reweighting

We employ a single 3-D convolution over (u, v, b) with kernel sizes $(p_h, p_w, p_s) \in \mathbb{N}^3$, strides $(s_h, s_w, s_s) \in \mathbb{N}^3$, padding mode Pad $\in$ {same, valid}, output width $d_{\text{emb}} \in 4\mathbb{N}$, and GELU activation. Concretely, $\phi_{3\text{D}}$ is a Conv3−D mapping one input channel to d_{emb} output channels

$$\mathbf{F}_n = \phi_{3\text{D}}\left(\widehat{\mathbf{P}}_n\right) \in \mathbb{R}^{H'\times W'\times S'\times d_{\text{emb}}}\ [3] \tag{1}$$

[1] Note: Unless stated otherwise, index ranges written $[i : i + L - 1]$ are inclusive here (half-open slices $[i : i + L)$ also supported).

[2] $\widehat{\mathbf{P}}_n(u, v, b, 1) \in \mathbb{R}^{W_S\times W_S\times B\times 1}$ is per-sample Z-score standardization.

[3] $(H', W', S') = \begin{cases} (\lceil W_S/s_h\rceil, \lceil W_S/s_w\rceil, \lceil B/s_s\rceil) \\ (\lfloor (W_S - p_h)/s_h\rfloor + 1, \lfloor (W_S - p_w)/s_w\rfloor + 1, \\ \lfloor (B - p_s)/s_s\rfloor + 1) \end{cases}$ for both same and valid padding, depending on the choice.

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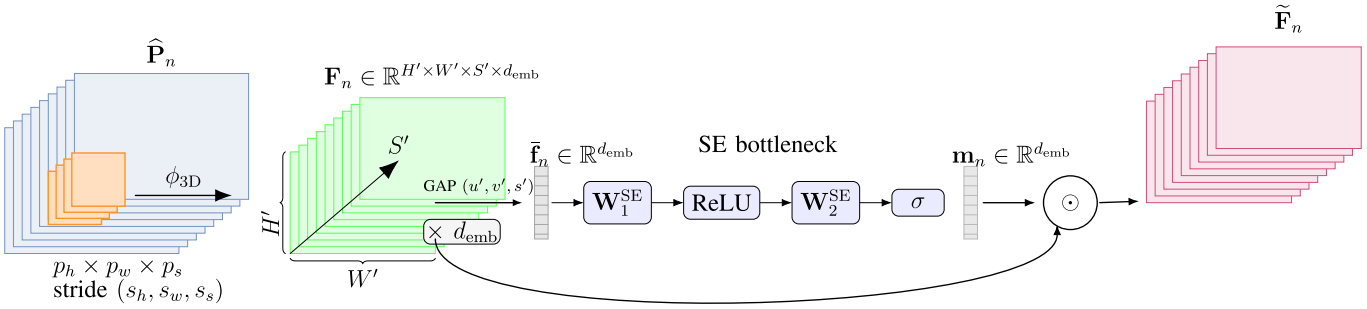


Fig. 1. 3-D patchifier and channel reweighting. A Conv3-D with kernel (p_h, p_w, p_s), stride (s_h, s_w, s_s), and d_{emb} output channels maps $\widehat{\mathbf{P}}_n$ to $\mathbf{F}_n$. Global average pooling over (u', v', s') yields $\bar{\mathbf{f}}_n$, which passes through an SE bottleneck $(\mathbf{W}_1^{\text{SE}}, \text{ReLU}, \mathbf{W}_2^{\text{SE}}, \sigma)$ to form gates $\mathbf{m}_n$; these reweight channels of $\mathbf{F}_n$ elementwise to produce $\widetilde{\mathbf{F}}_n$.

To adaptively emphasize informative channels, we apply SE over (u', v', s'). First, we compute a channel descriptor by global average pooling

$$\bar{\mathbf{f}}_n = \frac{1}{H'W'S'} \sum_{u'=1}^{H'} \sum_{v'=1}^{W'} \sum_{s'=1}^{S'} \mathbf{F}_n\left(u', v', s', :\right) \tag{2}$$

where $\bar{\mathbf{f}}_n \in \mathbb{R}^{d_{\text{emb}}}$. Then, a two-layer bottleneck MLP produces channel gates

$$\mathbf{m}_n = \sigma\left(\mathbf{W}_2^{\text{SE}} \text{ReLU}\left(\mathbf{W}_1^{\text{SE}} \bar{\mathbf{f}}_n\right)\right) \in \mathbb{R}^{d_{\text{emb}}} \tag{3}$$

with $\mathbf{W}_1^{\text{SE}} \in \mathbb{R}^{d_{\text{emb}} \times d_{\text{hid}}}$, $\mathbf{W}_2^{\text{SE}} \in \mathbb{R}^{d_{\text{hid}} \times d_{\text{emb}}}$, and $d_{\text{hid}} = \max\{8, \lfloor d_{\text{emb}}/4 \rfloor\}$. The gates are broadcast back to all spatial/spectral positions and applied elementwise

$$\widetilde{\mathbf{F}}_n\left(u', v', s', :\right) = \mathbf{F}_n\left(u', v', s', :\right) \odot \mathbf{m}_n \tag{4}$$

where $\widetilde{\mathbf{F}}_n(u', v', s', :) \in \mathbb{R}^{H' \times W' \times S' \times d_{\text{emb}}}$ and $\odot$ denotes the Hadamard (elementwise) product. The 3-D patchifier compacts local spatial–spectral neighborhoods into a uniform embedding grid while controlling resolution via (s_h, s_w, s_s). The complete process is presented in Fig. 1.

B. Band-Position Tensor and Spectral Downsampling

Let the normalized band coordinates be $\boldsymbol{\beta} \in [0, 1]^B$, set as $\beta_b = (b-1)/(B-1)$, $b \in \{1, \ldots, B\}$.[4] To align the band-coordinate length with the patchifier's spectral size S', we average-pool along the spectral axis using the same kernel and stride as the 3-D patchifier in the spectral dimension and the same padding mode $\text{Pad} \in \{\text{same}, \text{valid}\}$

$$\boldsymbol{\beta}^{\downarrow} = \text{Pool}_{1D}\left(\boldsymbol{\beta};\ k = p_s,\ t = s_s,\ \text{Pad}\right) \in \mathbb{R}^{S'} \tag{5}$$

Using physically meaningful band coordinates and downsampling them with (k, t, Pad) that mirror the 3-D patchifier guarantees $|\boldsymbol{\beta}^{\downarrow}| = S'$, enabling shape-consistent fusion with spectral features and improving robustness to nonuniform bands.

[4]If calibrated wavelengths $\boldsymbol{\lambda} \in \mathbb{R}^B$ are available, one may instead use $\beta_b = (\lambda_b - \min(\boldsymbol{\lambda}))/(\max(\boldsymbol{\lambda}) - \min(\boldsymbol{\lambda}) + \varepsilon)$ with a small $\varepsilon > 0$; both choices are supported.

C. Spectral Positional Encoding

We encode spectral locations by sinusoidal features at exponentially spaced frequencies. Given $F_{\text{sp}} \in \mathbb{N}$ and $d_{\text{pe}} \in \mathbb{N}$, for each $s' \in \{1, \ldots, S'\}$ defined as $\nu_f = 2^{f-1}$ where $f \in \{1, \ldots, F_{\text{sp}}\}$, and the $2F_{\text{sp}}$-dimensional embedding

$$\mathbf{e}\left(s'\right) = \left[\ \sin\left(2\pi\, \nu_f\, \beta_{s'}^{\downarrow}\right),\ \cos\left(2\pi\, \nu_f\, \beta_{s'}^{\downarrow}\right)\ \right]_{f=1}^{F_{\text{sp}}} \tag{6}$$

where $\mathbf{e}(s') \in \mathbb{R}^{2F_{\text{sp}}}$. A learned linear projection maps $\mathbf{e}(s')$ to a d_{pe}-vector

$$\mathbf{z}\left(s'\right) = \mathbf{W}_{\text{pe}}\, \mathbf{e}\left(s'\right) + \mathbf{b}_{\text{pe}} \in \mathbb{R}^{d_{\text{pe}}} \tag{7}$$

where $\mathbf{W}_{\text{pe}} \in \mathbb{R}^{d_{\text{pe}} \times 2F_{\text{sp}}}$, $\mathbf{b}_{\text{pe}} \in \mathbb{R}^{d_{\text{pe}}}$. After tiling $\mathbf{z}(s')$ uniformly over the spatial grid $(u', v') \in \{1, \ldots, H'\} \times \{1, \ldots, W'\}$, we obtain the spectral positional encoding (SPE) tensor

$$\mathbf{Z}^{\text{spe}}\left(u', v', s', :\right) = \mathbf{z}\left(s'\right) \in \mathbb{R}^{H' \times W' \times S' \times d_{\text{pe}}} \tag{8}$$

We then concatenate SPE with the SE-reweighted features $\widetilde{\mathbf{F}}_n$ along the channel axis and reduce back to width d_{emb} via a linear layer with GELU, followed by LayerNorm

$$\mathbf{G}_n\left(u', v', s', :\right) = \left[\ \widetilde{\mathbf{F}}_n\left(u', v', s', :\right) \| \mathbf{Z}^{\text{spe}}\left(u', v', s', :\right)\ \right] \tag{9}$$

$$\mathbf{H}_n = \text{LN}\left(\phi_{\text{lin}}\left(\mathbf{G}_n\right)\right) \in \mathbb{R}^{H' \times W' \times S' \times d_{\text{emb}}} \tag{10}$$

$$\phi_{\text{lin}} :\ \mathbb{R}^{d_{\text{emb}} + d_{\text{pe}}} \to \mathbb{R}^{d_{\text{emb}}} \text{ with GELU} \tag{11}$$

where $\mathbf{G}_n(u', v', s', :) \in \mathbb{R}^{H' \times W' \times S' \times (d_{\text{emb}} + d_{\text{pe}})}$. SPE injects explicit spectral-location information that is invariant to intensity, allowing subsequent attention to distinguish where in the spectrum a response occurs; the projection and normalization align these priors to the network's feature scale before ISA operates.

D. Irregular Spectral Aggregation

Let $R_{\text{ref}} \in \mathbb{N}$ be the number of spectral anchors. We learn raw anchor parameters $\mathbf{r} \in \mathbb{R}^{R_{\text{ref}}}$ and squash them to $[0, 1]$ via $\rho_r = \sigma(r_r)$ for $r \in \{1, \ldots, R_{\text{ref}}\}$. We first normalize the downsampled band coordinates to $[0, 1]$

$$\widehat{\beta}_{n,s'}^{\downarrow} = \frac{\beta_{s'}^{\downarrow} - \min\left(\boldsymbol{\beta}^{\downarrow}\right)}{\max\left(\boldsymbol{\beta}^{\downarrow}\right) - \min\left(\boldsymbol{\beta}^{\downarrow}\right) + \varepsilon}, \quad s' \in \left\{1, \ldots, S'\right\} \tag{12}$$

where $\varepsilon > 0$ is a small constant. The anchor–position dissimilarity and a shared MLP $\varphi(\cdot)$ produce logits

$$a_n^{\log}\left(r, s'\right) = \varphi\left(\Delta_n\left(r, s'\right)\right) \in \mathbb{R} \tag{13}$$

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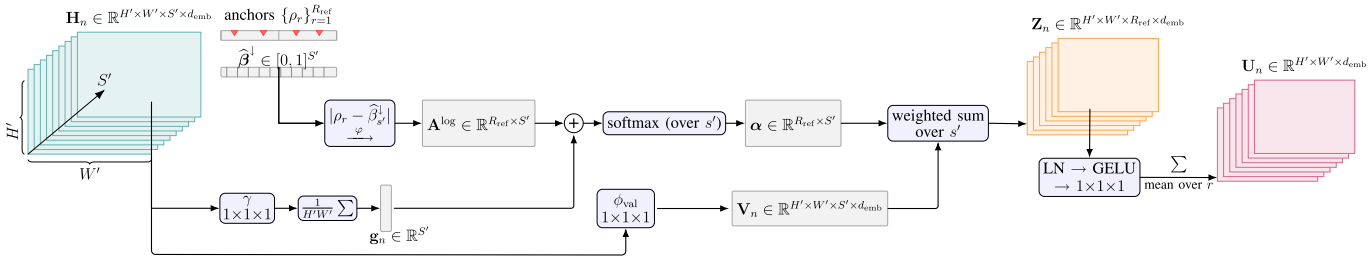


Fig. 2. ISA. Learned anchors $\{\rho_r\}$ on $[0,1]$ are compared with downsampled band positions $\widehat{\boldsymbol{\beta}}^{\downarrow}$. The dissimilarities are mapped by an MLP φ to logits and fused with a content gate $g_n(s')$ obtained from $\gamma(\mathbf{H}_n)$ followed by spatial averaging. Softmax over s' yields $\boldsymbol{\alpha}$, which aggregates values $\mathbf{V}_n = \phi_{\text{val}}(\mathbf{H}_n)$ along the spectral axis into $\mathbf{Z}_n$ (one slice per anchor). Post-ISA refinement (LN→GELU→ $1\times1\times1$) and averaging over anchors produce the spectrally condensed map $\mathbf{U}_n$.

where $\Delta_n(r,s') = \left|\rho_r - \widehat{\beta}^{\downarrow}_{n,s'}\right| \in \mathbb{R}$. A *content gate* is computed from the feature tensor $\mathbf{H}_n \in \mathbb{R}^{H'\times W'\times S'\times d_{\text{emb}}}$ via a $1\times1\times1$ convolution and spatial averaging

$$g_n\left(s'\right) = \frac{1}{H'W'}\sum_{u'=1}^{H'}\sum_{v'=1}^{W'}\mathbf{G}_n^{\text{raw}}\left(u',v',s',1\right)\in\mathbb{R} \tag{14}$$

where $\mathbf{G}_n^{\text{raw}} = \gamma(\mathbf{H}_n) \in \mathbb{R}^{H'\times W'\times S'\times 1}$. The final logits and attention weights (softmax over s') are

$$\alpha_n\left(r,s'\right) = \frac{\exp\left(\ell_n\left(r,s'\right)\right)}{\sum_{q=1}^{S'}\exp\left(\ell_n\left(r,q\right)\right)} \tag{15}$$

where $\ell_n(r,s') = a_n^{\log}(r,s') + g_n(s')$, with $\alpha_n \in \mathbb{R}^{R_{\text{ref}}\times S'}$. Values are obtained by a learned projection $\mathbf{V}_n = \phi_{\text{val}}(\mathbf{H}_n) \in \mathbb{R}^{H'\times W'\times S'\times d_{\text{emb}}}$ and aggregated at each spatial location (u',v')

$$\mathbf{Z}_n\left(u',v',r,:\right) = \sum_{s'=1}^{S'}\alpha_n\left(r,s'\right)\,\mathbf{V}_n\left(u',v',s',:\right)\in\mathbb{R}^{d_{\text{emb}}} \tag{16}$$

yielding $\mathbf{Z}_n \in \mathbb{R}^{H'\times W'\times R_{\text{ref}}\times d_{\text{emb}}}$. Later, we apply LayerNorm as $\bar{\mathbf{Z}}_n = \text{LN}(\mathbf{Z}_n)$ and two pointwise channel maps before collapsing the reference axis as $\tilde{\mathbf{Z}}_n = \text{GELU}(\bar{\mathbf{Z}}_n\,\mathbf{W}_d + \mathbf{b}_d)$ and $\check{\mathbf{Z}}_n = \phi_{1\times1\times1}(\tilde{\mathbf{Z}}_n)$, where $\mathbf{W}_d \in \mathbb{R}^{d_{\text{emb}}\times d_{\text{emb}}}$ acts channelwise and $\phi_{1\times1\times1}$ is a $1\times1\times1$ Conv3D with GELU. Finally, we average over anchors

$$\mathbf{U}_n\left(u',v',:\right) = \frac{1}{R_{\text{ref}}}\sum_{r=1}^{R_{\text{ref}}}\check{\mathbf{Z}}_n\left(u',v',r,:\right)\in\mathbb{R}^{d_{\text{emb}}} \tag{17}$$

so that $\mathbf{U}_n \in \mathbb{R}^{H'\times W'\times d_{\text{emb}}}$ is a spectrally condensed spatial feature map. ISA replaces a uniform, stationary spectral kernel with a small set of learned anchors that attend over bands in a content-aware manner. This does both: 1) reduce spectral tokens from S' to $R_{\text{ref}} \ll S'$ before spatial/token mixing (lowering compute) and 2) adapts to nonuniform band spacing or missing bands, improving robustness and discriminative power. The ISA process is defined in Fig. 2.

E. Irregular Spatial Propagation (ISA_{Spa})

Starting from the spectrally condensed feature map $\mathbf{U}_n \in \mathbb{R}^{H'\times W'\times d_{\text{emb}}}$, we define $N_{\text{sp}} = H'W'$ spatial tokens by row-major flattening as $\mathbf{X}_n^{\text{sp}} = \text{vec}(\mathbf{U}_n) \in \mathbb{R}^{N_{\text{sp}}\times d_{\text{emb}}}$. Let $\mathcal{C} = \{(y_k,x_k)\}_{k=1}^{N_{\text{sp}}}$ be the coordinates of the $H'\times W'$ grid (row-major order). For each pair (k,ℓ), form a normalized relative displacement

$$\boldsymbol{\delta}_{k\ell} = \frac{1}{\max\{H',W'\}}\begin{bmatrix}y_k - y_\ell\\ x_k - x_\ell\end{bmatrix}\in\mathbb{R}^2 \tag{18}$$

A shared two-layer MLP $\psi(\cdot)$ maps $\boldsymbol{\delta}_{k\ell}$ to a scalar logit $\eta_{k\ell}$, yielding a sample-independent affinity matrix

$$\eta_{k\ell} = \psi\left(\boldsymbol{\delta}_{k\ell}\right)\in\mathbb{R};\qquad \mathbf{E} = \left[\eta_{k\ell}\right]_{k,\ell=1}^{N_{\text{sp}}}\in\mathbb{R}^{N_{\text{sp}}\times N_{\text{sp}}} \tag{19}$$

Rowwise softmax converts logits to propagation weights

$$\pi_{k\ell} = \frac{\exp\left(\eta_{k\ell}\right)}{\sum_{q=1}^{N_{\text{sp}}}\exp\left(\eta_{kq}\right)};\qquad \boldsymbol{\Pi} = \left[\pi_{k\ell}\right]\in\mathbb{R}^{N_{\text{sp}}\times N_{\text{sp}}} \tag{20}$$

Applying the learned nonstationary kernel to features and projecting back to width d_{emb} as $\mathbf{Y}_n^{\text{sp}} = \boldsymbol{\Pi}\,\mathbf{X}_n^{\text{sp}}\mathbf{W}^{\text{out}} \in \mathbb{R}^{N_{\text{sp}}\times d_{\text{emb}}}$, where $\mathbf{W}^{\text{out}} \in \mathbb{R}^{d_{\text{emb}}\times d_{\text{emb}}}$. Finally, reshape back to the grid and normalize as $\widetilde{\mathbf{U}}_n = \text{reshape}(\mathbf{Y}_n^{\text{sp}}, H', W', d_{\text{emb}}) \in \mathbb{R}^{H'\times W'\times d_{\text{emb}}}$, where $\widetilde{\mathbf{U}}_n \leftarrow \text{LN}(\widetilde{\mathbf{U}}_n)$ and ISA_{spa} learns a geometry-driven, coordinate-kernel spatial propagation (via ψ on relative displacements) that propagates information anisotropically across the grid. Because $\boldsymbol{\Pi}$ depends only on $(y_k - y_\ell,\ x_k - x_\ell)$, it is lightweight and sample-independent, complementing the content-aware spectral attention that precedes it. The overall process is presented in Fig. 3.

F. Spatial Positional Encoding

We enrich $\widetilde{\mathbf{U}}_n \in \mathbb{R}^{H'\times W'\times d_{\text{emb}}}$ with: 1) a *fixed* 2-D sinusoidal encoding and 2) a *learnable* 2-D bias, which require $d_{\text{emb}} = 4d_q$ for $d_q \in \mathbb{N}$. **Fixed sinusoidal encoding:** Let rows $y \in \{1,\ldots,H'\}$ and columns $x \in \{1,\ldots,W'\}$. Define frequencies $\omega_{2m} = \exp(-2m/d_q \log 10\,000);\ m = 1,\ldots,d_q/2$. Form 1-D encodings for y and x

$$\mathbf{p}_y^{\sin}(y) = \left[\ \sin\left(\omega_{2m}y\right),\ \cos\left(\omega_{2m}y\right)\ \right]_{m=1}^{d_q/2}\in\mathbb{R}^{d_q} \tag{21}$$

$$\mathbf{p}_x^{\sin}(x) = \left[\ \sin\left(\omega_{2m}x\right),\ \cos\left(\omega_{2m}x\right)\ \right]_{m=1}^{d_q/2}\in\mathbb{R}^{d_q} \tag{22}$$

We interleave these into a $d_{\text{emb}} = 4d_q$-dimensional tensor $\mathbf{P}^{\sin} \in \mathbb{R}^{H'\times W'\times d_{\text{emb}}}$

$$\mathbf{P}^{\sin}(y,x,c) = \begin{cases}\left(\mathbf{p}_y^{\sin}(y)\right)_{2m-1}, & c = 2m-2\\ \left(\mathbf{p}_y^{\sin}(y)\right)_{2m}, & c = 2m-1\\ \left(\mathbf{p}_x^{\sin}(x)\right)_{2m-1}, & c = d_q+2m-2\\ \left(\mathbf{p}_x^{\sin}(x)\right)_{2m}, & c = d_q+2m-1\\ 0, & \text{otherwise}\end{cases} \tag{23}$$

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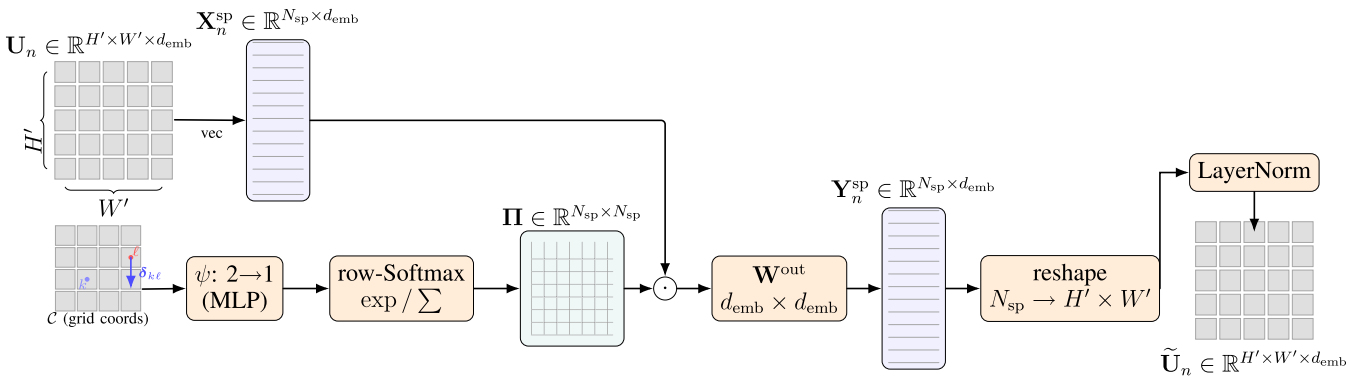


Fig. 3. Irregular spatial propagation (ISA$_{\text{spa}}$): vectorize $\mathbf{U}_n$ to tokens $\mathbf{X}_n^{\text{sp}}$, build sample-independent affinities from relative displacements via ψ and row-Softmax to obtain $\mathbf{\Pi}$, then propagate $\mathbf{\Pi}\mathbf{X}_n^{\text{sp}}\mathbf{W}^{\text{out}}$, reshape to the grid, and apply LayerNorm.

where $m = 1, \ldots, d_q/2$. This matches the requirement that d_{emb} be divisible by 4 and its channel assignment pattern. We add it elementwise

$$\mathbf{U}_n^{\text{sin}} = \widetilde{\mathbf{U}}_n + \mathbf{P}^{\text{sin}} \in \mathbb{R}^{H' \times W' \times d_{\text{emb}}} \tag{24}$$

1) Learnable 2-D Encoding: We introduce a trainable bias $\mathbf{P}^{\text{lrn}} \in \mathbb{R}^{1 \times H' \times W' \times d_{\text{emb}}}$ (broadcast over the batch)

$$\mathbf{U}_n^{\text{pos}} = \mathbf{U}_n^{\text{sin}} + \mathbf{P}^{\text{lrn}} \in \mathbb{R}^{H' \times W' \times d_{\text{emb}}} \tag{25}$$

The fixed term preserves absolute geometry at multiple spatial scales and is dataset-agnostic; the learnable term adapts to dataset-specific layout biases. Using the channel layout above ensures shape compatibility and avoids reparameterizing feature width.

G. Transformer Encoder

We flatten $\mathbf{U}_n^{\text{pos}} \in \mathbb{R}^{H' \times W' \times d_{\text{emb}}}$ into $N_{\text{sp}} = H'W'$ tokens and prepend a learned class token

$$\mathbf{Z}_n^{(0)} = \begin{bmatrix} \mathbf{z}_{\text{cls}} \\ \text{vec}\left(\mathbf{U}_n^{\text{pos}}\right) \end{bmatrix} \in \mathbb{R}^{(N_{\text{sp}}+1) \times d_{\text{emb}}} \tag{26}$$

We stack L encoder blocks. For layer $\ell \in \{1, \ldots, L\}$, the prenorm & iMHSA is defined as

$$\widehat{\mathbf{Z}}_n^{(\ell)} = \text{LN}\left(\mathbf{Z}_n^{(\ell-1)}\right) \tag{27}$$

$$\mathbf{A}_n^{(\ell)} = \text{iMHSA}\left(\widehat{\mathbf{Z}}_n^{(\ell)}\right) \in \mathbb{R}^{(N_{\text{sp}}+1) \times d_{\text{emb}}} \tag{28}$$

$$\mathbf{Z}_n^{(\ell)} = \mathbf{Z}_n^{(\ell-1)} + \text{DropPath}_{p_\ell}\left(\mathbf{\Gamma}_1 \odot \mathbf{A}_n^{(\ell)}\right) \tag{29}$$

where $\mathbf{\Gamma}_1 \in \mathbb{R}^{d_{\text{emb}}}$ is a LayerScale vector, p_ℓ is the stochastic-depth rate (linearly increased across layers), and $\odot$ denotes the Hadamard (elementwise) product. Prenorm and MLP blocks are defined as

$$\widetilde{\mathbf{Z}}_n^{(\ell)} = \text{LN}\left(\mathbf{Z}_n^{(\ell)}\right); \quad \mathbf{M}_n^{(\ell)} = \text{Drop}\left(\mathbf{W}_2^{(\ell)}\,\text{GELU}\left(\mathbf{W}_1^{(\ell)}\,\widetilde{\mathbf{Z}}_n^{(\ell)}\right)\right) \tag{30}$$

$$\mathbf{Z}_n^{(\ell)} \leftarrow \mathbf{Z}_n^{(\ell)} + \text{DropPath}_{p_\ell}\left(\mathbf{\Gamma}_2 \odot \mathbf{M}_n^{(\ell)}\right) \tag{31}$$

with $\mathbf{\Gamma}_2 \in \mathbb{R}^{d_{\text{emb}}}$ and dropout (Drop($\cdot$)) matching implementation.

1) Irregular MHSA: Let the number of heads be H_{mh}. We allow heterogeneous query/key channel sizes per head, $\{d_{qk}^{(h)}\}_{h=1}^{H_{\text{mh}}}$, and value sizes $\{d_v^{(h)}\}_{h=1}^{H_{\text{mh}}}$. Define $d_{\text{out}} = \sum_{h=1}^{H_{\text{mh}}} d_v^{(h)}$. For an input token matrix $\mathbf{T} \in \mathbb{R}^{(N_{\text{sp}}+1) \times d_{\text{emb}}}$, each head computes

$$\mathbf{Q}^{(h)} = \mathbf{T}\mathbf{W}_Q^{(h)} + \mathbf{b}_Q^{(h)} \in \mathbb{R}^{(N_{\text{sp}}+1) \times d_{qk}^{(h)}} \tag{32}$$

$$\mathbf{K}^{(h)} = \mathbf{T}\mathbf{W}_K^{(h)} + \mathbf{b}_K^{(h)} \in \mathbb{R}^{(N_{\text{sp}}+1) \times d_{qk}^{(h)}} \tag{33}$$

$$\mathbf{V}^{(h)} = \mathbf{T}\mathbf{W}_V^{(h)} + \mathbf{b}_V^{(h)} \in \mathbb{R}^{(N_{\text{sp}}+1) \times d_v^{(h)}} \tag{34}$$

Let $\mathbf{M} \in \{0, 1\}^{N \times N}$ be an attention mask (1 = keep and 0 = mask). With scaling and masking

$$\tilde{\mathbf{L}}^{(h)} = \frac{\mathbf{Q}^{(h)}\mathbf{K}^{(h)\top}}{\sqrt{d_{qk}^{(h)}}}; \quad \mathbf{L}^{(h)} = \tilde{\mathbf{L}}^{(h)} + (\mathbf{1} - \mathbf{M}) \cdot (-\infty) \tag{35}$$

Attention and attention dropout (p_{attn}) is $\mathbf{S}^{(h)} = \text{Softmax}(\mathbf{L}^{(h)})$ and $\hat{\mathbf{S}}^{(h)} = \text{Dropout}_{p_{\text{attn}}}(\mathbf{S}^{(h)})$ with $\mathbf{C}^{(h)} = \hat{\mathbf{S}}^{(h)}\mathbf{V}^{(h)} \in \mathbb{R}^{N \times d_v^{(h)}}$. Concatenate and project (let $d_{\text{out}} = \sum_h d_v^{(h)}$ and $d_{\text{proj}} = \text{output_dim}$)

$$\mathbf{H} = \left[\, \mathbf{C}^{(1)} \,\|\, \cdots \,\|\, \mathbf{C}^{(H_{\text{mh}})} \,\right] \in \mathbb{R}^{N \times d_{\text{out}}} \tag{36}$$

$$\mathbf{Y} = \mathbf{H}\,\mathbf{W}_O + \mathbf{b}_O, \quad \mathbf{W}_O \in \mathbb{R}^{d_{\text{out}} \times d_{\text{proj}}} \tag{37}$$

Finally, apply projection dropout (p_{proj}) as iMHSA($\mathbf{T}$) = $\text{Dropout}_{p_{\text{proj}}}(\mathbf{Y}) \in \mathbb{R}^{N \times d_{\text{proj}}}$, allowing per-head query/key sizes $\{d_{qk}^{(h)}\}$ makes the attention irregular, letting different heads specialize in heterogeneous subspaces while keeping the concatenated value width controlled by $\{d_v^{(h)}\}$. The prenorm, LayerScale, and DropPath scheme stabilizes deep stacking and improves optimization robustness. The process is presented in Figs. 4 and 5.

H. Classifier

Let $\mathbf{Z}_n^{(L)} \in \mathbb{R}^{(N_{\text{sp}}+1) \times d_{\text{emb}}}$ be the final transformer output for sample n. We extract the class token and the average of the remaining tokens

$$\mathbf{c}_n = \mathbf{Z}_n^{(L)}[1, :] \in \mathbb{R}^{d_{\text{emb}}} \tag{38}$$

$$\mathbf{g}_n = \frac{1}{N_{\text{sp}}} \sum_{k=2}^{N_{\text{sp}}+1} \mathbf{Z}_n^{(L)}[k, :] \in \mathbb{R}^{d_{\text{emb}}} \tag{39}$$

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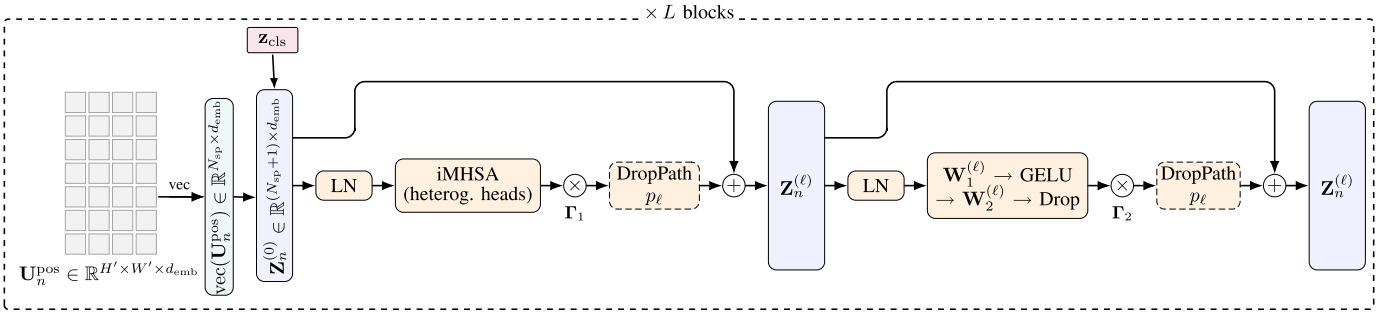


Fig. 4. Transformer encoder with prenorm, irregular MHSA (see Fig. 5), LayerScale $\mathbf{\Gamma}_1, \mathbf{\Gamma}_2$, and DropPath p_ℓ on each residual branch. Tokens are formed by prepending a learned class token to $\mathrm{vec}(\mathbf{U}_n^{\mathrm{pos}})$.

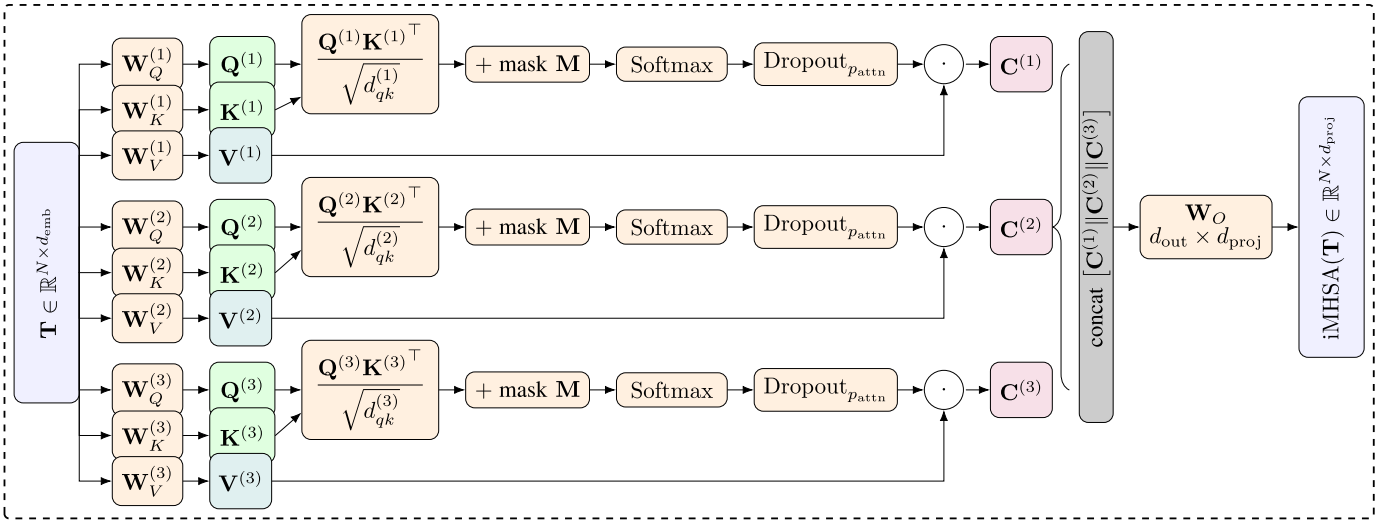


Fig. 5. IMHSA with heterogeneous per-head query/key and value widths, including masking and attention dropout. For each head: scaled logits $(\mathbf{Q}^{(h)}\mathbf{K}^{(h)\top}/(d_{qk}^{(h)})^{1/2})$ are masked by $\mathbf{M}$, softmaxed, then passed through $\mathrm{Dropout}_{p_{\mathrm{attn}}}$ before multiplying by $\mathbf{V}^{(h)}$ to form $\mathbf{C}^{(h)}$. Head contexts are concatenated and projected by $\mathbf{W}_O \in \mathbb{R}^{d_{\mathrm{out}} \times d_{\mathrm{proj}}}$ to yield $\mathrm{iMHSA}(\mathbf{T}) \in \mathbb{R}^{N \times d_{\mathrm{proj}}}$.

We concatenate and normalize to form the prelogit vector

$$\mathbf{h}_n = \mathrm{LN}\left(\left[\mathbf{c}_n \| \mathbf{g}_n\right]\right) \in \mathbb{R}^{2d_{\mathrm{emb}}} \qquad \mathbf{h}_n \leftarrow \mathrm{Drop}_{p_{\mathrm{rd}}}(\mathbf{h}_n) \tag{40}$$

The classifier is a linear map to C_{eff} classes followed by a softmax

$$\widehat{\mathbf{p}}_n = \mathrm{Softmax}\left(\mathbf{W}_{\mathrm{cls}}\mathbf{h}_n + \mathbf{b}_{\mathrm{cls}}\right) \in \Delta^{C_{\mathrm{eff}}-1} \tag{41}$$

with $\mathbf{W}_{\mathrm{cls}} \in \mathbb{R}^{C_{\mathrm{eff}} \times 2d_{\mathrm{emb}}}$ and $\mathbf{b}_{\mathrm{cls}} \in \mathbb{R}^{C_{\mathrm{eff}}}$. Concatenating the global class token $\mathbf{c}_n$ with the token-average $\mathbf{g}_n$ couples holistic context with distributed evidence over the spatial grid, improving calibration and robustness before the final linear decision layer.

III. Experimental Datasets and Settings

Iformer is evaluated on four widely used HSI benchmarks, namely, WHU-Hi [HongHu (HH) and HanChuan (HC)], Salinas (SA), and QUH-Tangdaowan (TG), covering agricultural and urban scenes with diverse spatial resolutions and spectral characteristics. For all datasets, the fixed-size spatial windows are extracted around labeled pixels using dense sliding windows over reflect-padded images, producing patch tensors $\mathbf{P} \in \mathbb{R}^{N \times W_S \times W_S \times B \times 1}$, as described in Section II. Unless otherwise stated, only foreground samples are retained during training and evaluation. Train/validation/test splits follow the standard protocols adopted in prior HSIC studies, and all reported results for Iformer are averaged over five independent runs with different random seeds. To avoid information leakage, normalization is applied independently to each sample using only patch-local statistics, without using information from validation and test data during training. Performance is evaluated using overall accuracy (OA), average accuracy (AA), and Cohen's κ coefficient.

The proposed Iformer employs a 3-D patchifier and SE-based embedding pipeline with embedding width $d_{\mathrm{emb}} = 64$, $L = 4$ encoder blocks, heterogeneous attention head dimensions $(12, 16, 20, 16)$, value width $d_v = 32$, spectral anchors $R_{\mathrm{ref}} = 8$, and SPE width $d_{\mathrm{pe}} = 32$. The 3-D patchifier uses kernel sizes $(p_h, p_w, p_s) = (4, 4, 8)$ with strides $(s_h, s_w, s_s) = (2, 2, 8)$ and same padding. Inputs are standardized using per-sample, per-band Z-score normalization together with reflection padding. When calibrated wavelengths are unavailable, the spectral coordinates are assumed to be uniformly spaced and are downsampled to match the spectral stride of the patchifier. Optimization uses AdamW with a warmup-cosine learning-rate schedule, base learning rate 5×10^{-4}, weight decay 2×10^{-4}, gradient clipping (1.0), EMA updates, dropout (0.1), and stochastic depth increasing linearly

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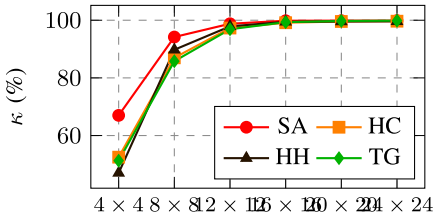


Fig. 6. Performance across several patch sizes.

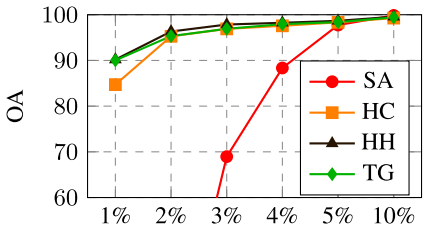


Fig. 7. Performance across several training samples.

from 0 to 0.1 across layers. Label smoothing ($\alpha = 0.05$), inverse-frequency class weighting, early stopping, and checkpoint selection based on validation accuracy are additionally employed to improve optimization stability and robustness under class imbalance.

IV. Parameter and Robustness Analysis

A. Patch Size Sensitivity

Fig. 6 illustrates the sensitivity of Iformer to the spatial window size across four benchmark datasets. Performance improves substantially when increasing the patch size from small windows (e.g., 4×4 or 8×8), indicating that larger neighborhoods provide richer spatial context for discrimination. Beyond moderate patch sizes, the performance gradually saturates, suggesting diminishing returns from additional context. Very large windows may introduce redundant or less informative spatial regions without significantly improving classification accuracy. Across all datasets, a patch size of 16×16 consistently lies within the stable high-performance region, providing a practical balance between contextual coverage and computational efficiency. Therefore, 16×16 is adopted as the default spatial window size in all subsequent experiments.

B. Training Sample Sensitivity

Fig. 7 evaluates the influence of training sample availability on classification performance across four benchmark datasets. As expected, performance improves progressively as the percentage of labeled training samples increases. Notably, the proposed Iformer maintains relatively strong performance even under highly limited supervision (e.g., 1%–3% training data), suggesting improved data efficiency in low-sample regimes. The performance gain becomes more gradual beyond 5% training data, indicating that the proposed irregular spatial–spectral modeling is able to capture representative structure with comparatively limited supervision.

C. T-SNE Visualization

To further assess the discriminative quality of the learned feature representations, we visualize the final patch-level

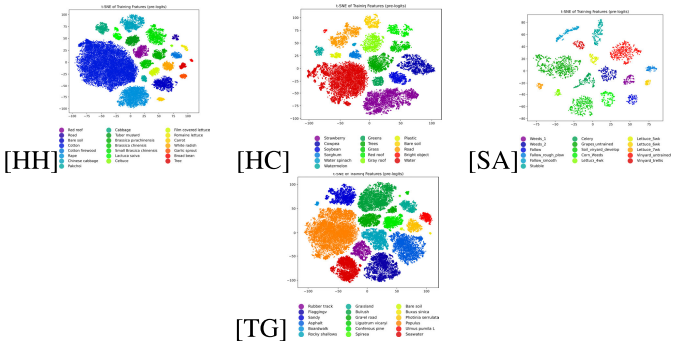


Fig. 8. t-SNE visualization of the learned embeddings. Each color corresponds to a ground-truth class.

TABLE I

Robustness Analysis of the Proposed Iformer Under Spectral Disorder and Boundary Sensitivity Conditions

Setting / Component	Spectral disorder (band-shuffle)				Boundary sensitivity	
	OA (%)	AA (%)	κ (%)	Δ	Mean F1 (%)	Δ
Full model	**98.6**	**98.1**	**98.5**	–	**95.5**	–
–SPE (no SPE)	96.1	95.7	96.0	-3.3	92.9	-2.6
–ISA (no ISA)	96.2	95.8	96.1	-3.2	93.1	-2.4
–ISA_spa (no ISA_spa)	97.4	97.0	97.3	-1.2	92.1	-3.4
Depthwise 3×3 conv (replace ISA_spa)	97.5	97.1	97.4	-1.1	93.4	-2.1
Std off (no per-sample Z-score)	97.8	97.3	97.6	-0.9	94.0	-1.5

embeddings using t-SNE on four representative datasets. As shown in Fig. 8, the learned embeddings form compact intraclass clusters with relatively clear interclass separation, including for spectrally similar materials. These observations suggest that the proposed irregular spectral and spatial modeling improves feature discrimination and representation consistency across heterogeneous scenes.

D. Robustness Analysis

Table I evaluates the robustness of Iformer under two stress conditions designed to probe spectral and spatial sensitivity. The first condition introduces spectral disorder by randomly permuting spectral bands at test time while preserving band values, thereby disrupting the original wavelength ordering. The second condition evaluates boundary sensitivity using boundary-F1 computed within a narrow trimap around class boundaries. Under band-order shuffling, the complete model maintains over 98% OA, whereas removing SPE or ISA produces consistent degradation in OA and κ. These results suggest that explicit spectral coordination and anchor-based aggregation improve robustness to irregular or perturbed spectral organization. For boundary-sensitive evaluation, removing ISAspa reduces boundary-F1 by approximately 3–4,pt, indicating weaker structural consistency near class transitions. Replacing ISAspa with a depthwise 3×3 convolution recovers only part of this loss, suggesting that coordinate-aware propagation better captures anisotropic spatial relationships than isotropic local filtering.

E. Model Scale and Token Budget

Table II analyzes the tradeoff between model capacity, token density, and computational cost on the HH dataset, with similar trends observed on the remaining benchmarks. Increasing the embedding width (d_{emb}) or the number of encoder layers (L)

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TABLE II

MODEL SCALE AND TOKEN-BUDGET SWEEPS. H' AND W' ARE DETERMINED BY (p_h, p_w, s_h, s_w), AND S' IS DETERMINED BY (p_s, s_s)

d_{emb}	L	(p_h, p_w, p_s)	(s_h, s_w, s_s)	R_{ref}	$H' \times W' \times S'$/**N**	**OA**	**Train/Test (s)**
32	2	(4,4,8)	(2,2,8)	8	$10 \times 10 \times 2/100$	99.38	5420/130
64	4	(4,4,8)	(2,2,8)	8	$10 \times 10 \times 2/100$	**99.55**	6171/150
96	4	(4,4,8)	(2,2,8)	8	$10 \times 10 \times 2/100$	99.57	7320/168
64	6	(4,4,8)	(2,2,8)	8	$10 \times 10 \times 2/100$	99.58	6980/165
64	4	(4,4,8)	(1,1,8)	8	$20 \times 20 \times 2/400$	99.58	8010/181
64	4	(4,4,4)	(2,2,4)	8	$10 \times 10 \times 4/100$	99.56	6540/156
64	4	(4,4,12)	(2,2,12)	8	$10 \times 10 \times 2/100$	99.52	5980/147
64	4	(4,4,8)	(2,2,8)	4	$10 \times 10 \times 2/100$	99.43	6090/149
64	4	(4,4,8)	(2,2,8)	16	$10 \times 10 \times 2/100$	99.56	6620/154

TABLE III

INFLUENCE OF HETEROGENEOUS HEAD DIMENSIONS, VALUE-CHANNEL WIDTHS, AND SPE CONFIGURATIONS ON THE HH DATASET

Setting	**Head QK dims**	d_v	d_{pe}	**SPE freqs**	**OA (%)**	**AA (%)**
Baseline (irregular heads)	(12,16,20,16)	32	32	8	**99.55**	**99.30**
Equal heads (no specialization)	(16,16,16,16)	32	32	8	98.9	98.7
More heterogeneous	(8,16,24,24)	32	32	8	99.6	99.3
Wider value channels	(12,16,20,16)	48	32	8	99.4	99.1
Larger SPE width	(12,16,20,16)	32	64	8	99.5	99.2
Smaller SPE width	(12,16,20,16)	32	16	8	98.8	98.5
More SPE frequencies	(12,16,20,16)	32	32	12	99.6	99.3
Fewer SPE frequencies	(12,16,20,16)	32	32	4	98.9	98.6

provides only marginal performance gains while increasing computational cost, suggesting that the default configuration ($d_{\text{emb}} = 64$, $L = 4$) provides an effective balance between complexity and accuracy. Reducing spatial strides increases the number of spatial tokens substantially and leads to higher training and inference costs, while producing only limited improvements in OA. Similarly, varying the spectral stride changes the number of spectral tokens S', affecting the trade-off between contextual coverage and efficiency. The number of spectral anchors R_{ref} also influences this balance: fewer anchors reduce representation capacity, whereas larger anchor counts provide minimal additional benefit at increased computational cost.

F. Specialization Sweeps

Table III investigates the influence of heterogeneous attention-head allocation and SPE configurations. The default irregular-head configuration $(12, 16, 20, 16)$ achieves the strongest overall performance, whereas equal-width heads produce a noticeable reduction in OA and AA. This behavior suggests that heterogeneous attention allocation improves specialization across complementary spatial–spectral subspaces. Increasing the variability of head dimensions and widening the value channels produces only modest gains, indicating that the default setting already provides sufficient representational capacity. Similarly, the SPE configurations show that larger embedding widths and additional sinusoidal frequencies slightly improve performance, whereas reducing either parameter leads to more noticeable degradation. These observations indicate that sufficiently expressive SPE contributes to stable wavelength-aware feature modeling.

V. COMPONENT-LEVEL ABLATION

Table IV presents the componentwise ablations designed to isolate the contribution of the principal modules of Iformer while keeping the preprocessing pipeline, optimization settings, and training protocol fixed. Removing major modules, such as ISA, SPE, or ISAspa, consistently produces noticeable reductions in OA, AA, and κ across all datasets, indicating that both ISA and geometry-aware spatial propagation contribute meaningfully to the final performance. In particular, replacing ISA with uniform spectral averaging results in approximately 2–3,pt degradation, suggesting that the gains arise from coordinate-aware spectral modeling rather than from spectral compression alone. Similarly, replacing ISAspa with a standard depthwise convolution only partially recovers performance, indicating that anisotropic coordinate-driven propagation captures structural dependencies beyond isotropic local filtering. In contrast, auxiliary refinements, such as EMA updates, warmup scheduling, and label smoothing, produce comparatively smaller changes, suggesting that they primarily improve optimization stability rather than serving as the main source of performance gains.

VI. COMPARATIVE RESULTS

To evaluate the effectiveness of the proposed Iformer, we compare it with representative HSIC approaches spanning three major categories: 1) convolutional networks, including AGCNN [52] and SCSNet [53]; 2) transformer-based models, including BiFormer [54], HViT [55], SSFormer [36], WaveFormer [56], and MFormer [57]; and 3) state-space and Mamba-based architectures, including SSMamba [58], WaveMamba [59], MorpMamba [23], GraphMamba [60], and PMamba [1].

Table V and Figs. 9–12 summarize the comparative results. Across all datasets, Iformer achieves the strongest reported OA, AA, and Cohen's κ, indicating that the proposed design is competitive with, and in many cases more effective than, the compared methods. In particular, the improvements are consistent across datasets rather than concentrated on a single benchmark, which suggests that the gains are not tied to one specific scene and sensor configuration. The results also indicate that the combination of ISA and geometry-aware spatial propagation provides a useful inductive bias for HSIC under diverse operating conditions.

The convolutional baselines AGCNN and SCSNet provide strong local discrimination, but their performance is generally lower than Iformer, especially on HH and HC, where the scenes contain substantial spectral variability and fine-grained class structure. On these datasets, Iformer attains 99.56% and 99.26% OA, respectively, compared with mid-90% accuracy for the convolutional methods. This gap suggests that explicit spectral anchoring and adaptive band weighting may better accommodate nonstationary spectral responses than fixed convolutional kernels alone. On SA, where the overall benchmark is more saturated, the margin is smaller but remains consistent, indicating that the proposed model retains an advantage even in easier settings.

The transformer-based models, generally improve over CNNs by capturing longer range dependencies. However, their use of uniform tokenization and standard attention mechanisms may limit flexibility when the spectral axis exhibits irregular structure or when spatial patterns are highly heterogeneous. As summarized in Table V, Iformer remains competitive on all four datasets, while also using a compact parameter budget. In this setting, the ISA module reduces

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TABLE IV

COMPONENT ABLATIONS ON FOUR BENCHMARKS. Δ DENOTES THE ABSOLUTE OA DROP (%) RELATIVE TO THE FULL MODEL

Ablation	HH				HC				SA				TG			
	OA	Δ	AA	κ	OA	Δ	AA	κ	OA	Δ	AA	κ	OA	Δ	AA	κ
gray!12 **Full model baseline**																
Full model	**99.55**	–	**99.30**	**99.44**	**99.26**	–	**98.74**	**99.13**	**99.84**	–	**99.87**	**99.82**	**99.41**	–	**99.22**	**99.33**
gray!12 **Core methodological ablations**																
ISA_spa off	97.0	-2.5	96.6	96.9	96.8	-2.5	96.4	96.7	97.8	-2.0	97.4	97.7	97.2	-2.2	96.8	97.1
ISA_spa → depthwise 3×3 conv	97.9	-1.6	97.5	97.8	97.6	-1.7	97.2	97.5	98.5	-1.3	98.1	98.4	97.7	-1.7	97.3	97.6
SPE off	97.2	-2.3	96.8	97.1	97.0	-2.3	96.5	96.9	97.9	-1.9	97.5	97.8	97.3	-2.1	96.8	97.1
ISA → mean pooling	96.9	-2.6	96.4	96.8	96.7	-2.6	96.3	96.6	97.6	-2.2	97.2	97.5	97.0	-2.4	96.5	96.8
ISA without gate	97.9	-1.6	97.5	97.8	97.6	-1.7	97.2	97.5	98.5	-1.3	98.1	98.4	97.7	-1.7	97.3	97.6
iMHSA → equal head dims	98.2	-1.3	97.8	98.1	97.9	-1.4	97.5	97.8	98.7	-1.1	98.4	98.6	97.8	-1.6	97.4	97.7
$R_{\text{ref}} = 4$	98.2	-1.3	97.8	98.1	97.9	-1.4	97.5	97.8	98.7	-1.1	98.4	98.6	97.8	-1.6	97.4	97.7
$R_{\text{ref}} = 16$	99.6	+0.1	99.3	99.5	99.3	+0.0	98.8	99.1	99.8	+0.0	99.9	99.8	99.4	+0.0	99.2	99.3
gray!12 **Supporting architectural refinements**																
SE off	98.3	-1.2	97.8	98.2	98.0	-1.3	97.6	97.9	98.8	-1.0	98.5	98.7	97.9	-1.5	97.5	97.8
No fixed 2D PE	98.5	-1.0	98.1	98.4	98.2	-1.1	97.8	98.1	98.9	-0.9	98.6	98.8	98.1	-1.3	97.7	98.0
No learnable 2D PE	98.4	-1.1	98.0	98.3	98.1	-1.2	97.7	98.0	98.9	-0.9	98.6	98.8	98.0	-1.4	97.6	97.9
gray!12 **Optimization / engineering refinements**																
No class weights / label smoothing	98.5	-1.0	98.1	98.4	98.2	-1.1	97.8	98.1	98.9	-0.9	98.6	98.8	98.1	-1.3	97.7	98.0
EMA off	98.8	-0.7	98.4	98.7	98.5	-0.8	98.1	98.4	99.1	-0.7	98.8	99.0	98.6	-0.8	98.2	98.5
Constant LR	98.8	-0.7	98.4	98.7	98.5	-0.8	98.1	98.4	99.1	-0.7	98.8	99.0	98.6	-0.8	98.2	98.5

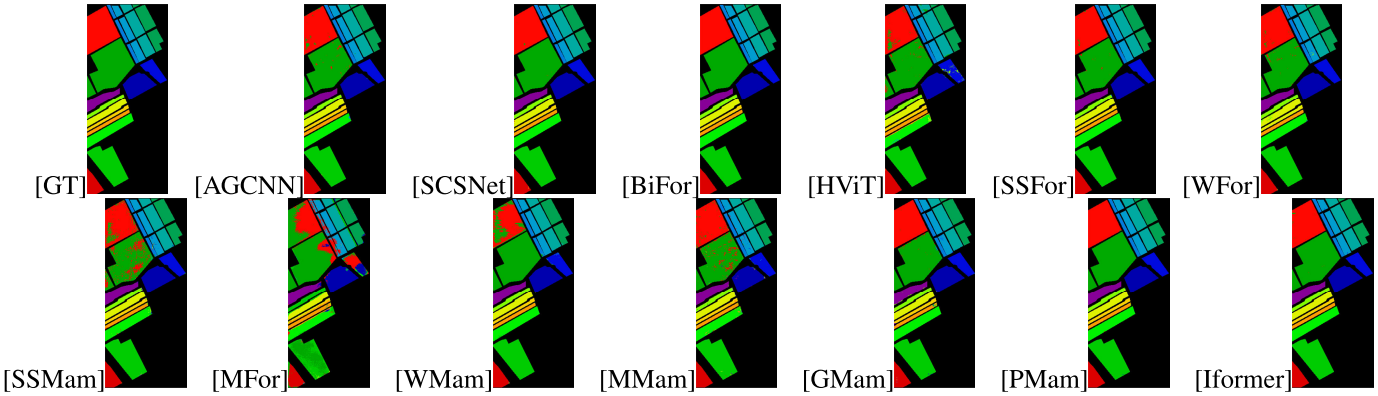


Fig. 9. Comparison of classification maps on the SA dataset.

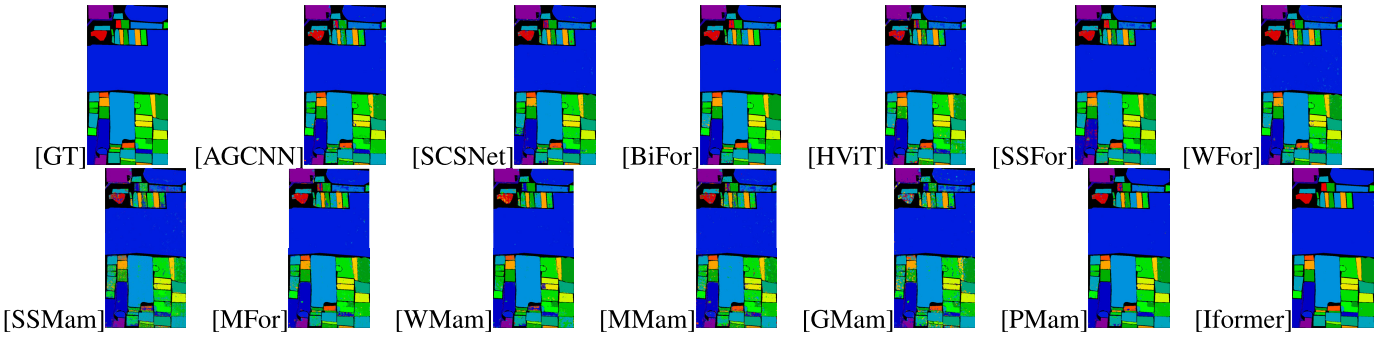


Fig. 10. Comparison of classification maps on the HH dataset.

spectral tokens before subsequent mixing, which helps control complexity, and iMHSA allows heterogeneous head dimensions that may better distribute representational capacity across complementary subspaces. These design choices appear to be particularly helpful when the data contain mixed pixels, narrow objects, or varying spectral signatures.

The recent Mamba-based methods, provide another strong comparison point because they are designed to model long-range dependencies efficiently. Their performance, however, varies across datasets, which suggests that sequential state propagation may be sensitive to scene-specific spectral ordering and distribution shifts. In contrast, Iformer maintains consistently strong results, which indicates a more stable behavior across different benchmark characteristics. This observation is consistent with the interpretation that the proposed irregular spectral–spatial modeling better accommodates nonuniform spectral organization and anisotropic spatial context.

The classification maps in Figs. 9–12 qualitatively support the quantitative results. Compared with the baseline methods, Iformer generally produces cleaner class boundaries and fewer fragmented regions, especially in areas where

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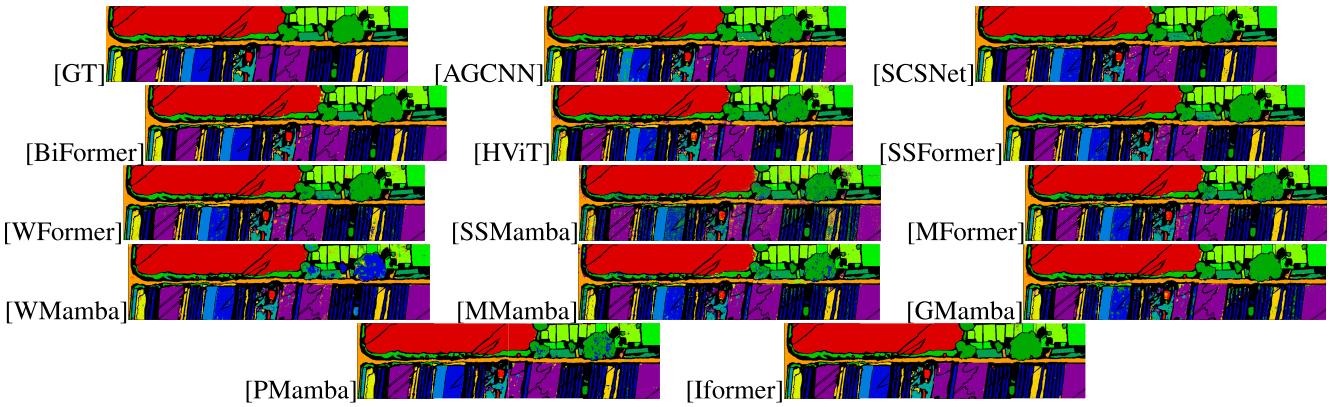


Fig. 11. Comparison of classification maps on the HC dataset.

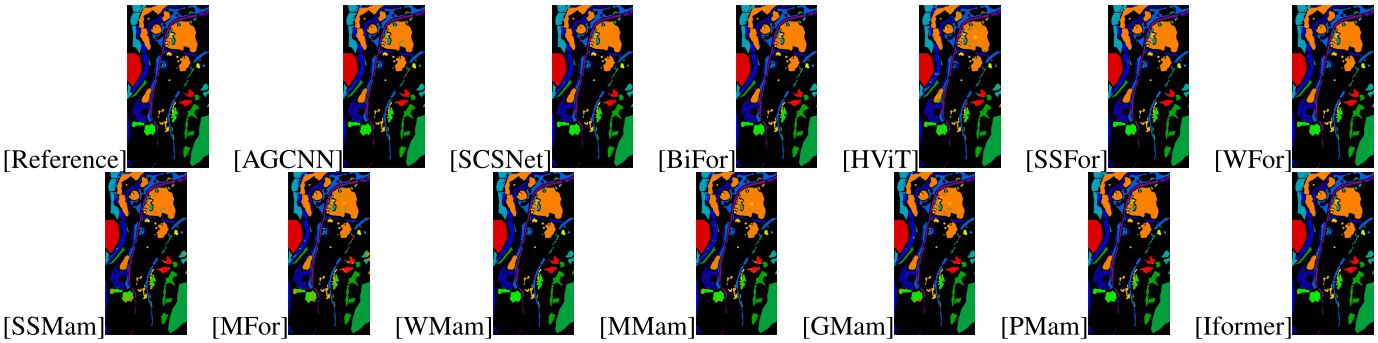


Fig. 12. Comparison of classification maps on the TG dataset.

TABLE V
COMPARATIVE PERFORMANCE OF IFORMER AGAINST THE STATE-OF-THE-ART HSIC MODELS ON FOUR BENCHMARK DATASETS

Metric	AGCNN	SCSNet	BiFormer	HViT	SSFormer	WaveFormer	SSMamba	MFormer	WaveMamba	MorpMamba	GraphMamba	PMamba	Iformer
HH dataset													
κ	94.6256	95.0855	97.3691	91.7616	94.2075	93.9063	80.4217	91.1545	90.9155	88.0689	83.9043	96.7382	**99.44±0.11**
OA	95.7472	96.1154	97.9192	93.4974	95.4279	95.1728	84.5843	93.0128	92.8056	90.5723	87.3329	97.4201	**99.56±0.10**
AA	87.9257	89.4886	94.5404	85.2938	91.0031	89.4710	65.5868	84.7170	82.4471	79.1826	66.8881	91.8958	**99.31±0.14**
HC dataset													
κ	95.9389	95.2453	97.0076	92.7191	94.5828	94.0240	78.8423	90.1047	87.9009	90.4156	92.3574	95.5058	**99.13±0.12**
OA	96.5309	95.9383	97.4420	93.7842	95.3738	94.8962	81.9608	91.5441	89.6390	91.8184	93.4677	96.1616	**99.26±0.11**
AA	92.7893	91.7600	95.1267	86.1313	89.8918	90.7267	66.0643	84.4788	85.8175	81.1019	86.6117	92.2893	**98.74±0.16**
SA dataset													
κ	99.0125	99.4508	99.6811	96.8106	99.5654	97.1246	93.0755	68.0344	94.6243	96.1814	98.5296	99.7454	**99.83±0.05**
OA	99.1132	99.2660	99.7136	97.1365	99.6097	97.4182	93.7719	71.8270	95.1898	96.5661	98.6791	99.7713	**99.85±0.04**
AA	99.5075	99.5749	99.3138	97.4169	99.6642	93.3160	96.6640	64.4138	97.0778	96.8948	98.1476	99.6014	**99.88±0.06**
TG dataset													
κ	96.9822	98.3745	98.5079	92.6695	98.4805	98.0710	93.4463	91.5298	98.0448	94.3556	94.9343	97.8315	**99.34±0.09**
OA	97.3473	98.5720	98.6902	93.5441	98.6647	98.3045	94.2578	92.5908	98.2816	95.0528	95.5543	98.0970	**99.42±0.08**
AA	95.6200	97.4177	96.7368	77.9612	97.8020	97.5266	91.5157	81.3397	97.0746	89.8162	85.8130	96.7378	**99.23±0.11**

spectral classes are visually similar or spatially intertwined. This is consistent with the role of ISA in separating spectral responses and ISA_{spa} in preserving directional spatial structure.

A. Computational Efficiency and Scalability

We further examine the computational characteristics of Iformer on the SA dataset. For the default configuration ($d_{\text{emb}} = 64$, $L = 4$, and $R_{\text{ref}} = 8$), the model achieves a favorable balance between accuracy and computational cost, requiring 705 s for training and 22 s for inference. Fig. 13 compares the inference time and parameter count across representative baseline models. Fig. 13 indicates that Iformer occupies a competitive region in the accuracy-efficiency space, with a compact parameter footprint and moderate inference latency.

Compared with transformer-based models, Iformer uses fewer parameters while remaining competitive in accuracy. Compared with several Mamba-based models, it avoids the large inference cost observed in some sequence-oriented architectures while preserving strong predictive performance. These results suggest that the proposed irregular spatial–spectral modeling is not only effective for representation learning but also practical from a computational standpoint. In this

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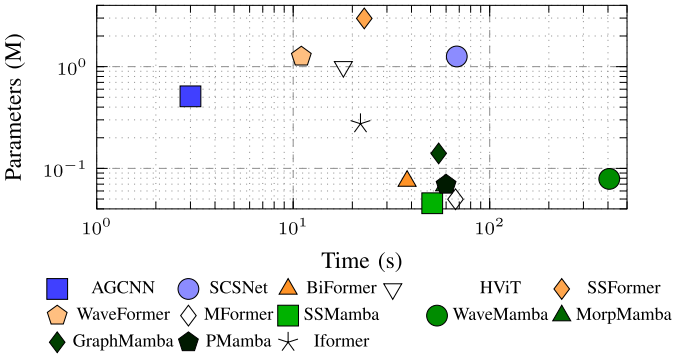


Fig. 13. Cross-model comparison of inference time and model size on the SA dataset.

sense, Iformer offers a favorable tradeoff between accuracy, efficiency, and model size for HSIC.

VII. Conclusion and Future Research Directions

This article presented Iformer, a band-aware irregular spatial–spectral transformer designed to address several persistent challenges in hyperspectral image classification, including spectral nonstationarity, anisotropic spatial structure, class imbalance, and the computational limitations of conventional CNN- and transformer-based architectures. The proposed framework combines coordinate-aware spectral modeling with two complementary aggregation mechanisms, namely, ISA and irregular spatial propagation (ISA_{spa}), to better capture irregular spectral geometry and direction-dependent spatial context. Together with the 3-D patchifier, SE, SPE, heterogeneous MHSA (iMHSA), and the calibrated CLS+GAP readout, the resulting architecture provides a unified spatial–spectral modeling pipeline that is effective while remaining computationally efficient.

Several directions remain open for future work. One promising avenue is dynamic spectral adaptation, in which spectral anchors could be conditioned on sensor metadata or wavelength uncertainty to improve generalization to unseen band configurations. Another direction is extending the irregular aggregation principle to multimodal and temporal settings, such as LiDAR, SAR, multispectral imagery, or time-series remote sensing data, to support joint spectral–spatial–temporal reasoning for tasks, such as crop monitoring, environmental analysis, and change detection. A further line of research is physics-aware interpretability, where radiative transfer or material reflectance priors may be incorporated into spectral anchor learning to produce more physically meaningful and explainable representations.

References

[1] M. Ahmad, M. Mazzara, S. Distefano, A. Mehmood Khan, M. Hassaan Farooq Butt, and D. Hong, "PolicyMamba: Localized policy attention with state space model for land cover classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 17814–17825, Oct. 2025.

[2] Q. Ren, M. Xu, N. Li, W. He, and S. Jia, "DiLAST: Leveraging differential RGB features for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5525015.

[3] M. Ahmad et al., "A comprehensive survey for hyperspectral image classification: The evolution from conventional to transformers and mamba models," *Neurocomputing*, vol. 644, Sep. 2025, Art. no. 130428.

[4] Y. Yuan, Q. Zang, Z. Lin, J. Liu, and Z. Fan, "Hierarchical pixel-wavelength fusion network for progressive hyperspectral image reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4409816.

[5] N. Yang, M. Fan, W. Wang, and H. Zhang, "Decision-making large language model for wireless communication: A comprehensive survey on key techniques," *IEEE Commun. Surveys Tuts.*, vol. 28, pp. 3055–3088, 2026.

[6] Y. Zhou, B. Li, H.-M. Hu, X. Zhang, D. Zhang, and H. Wang, "Heterogeneous feature re-sampling for balanced pedestrian attribute recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2706–2722, Apr. 2025.

[7] Y. Luo, J. Chen, R. Cong, H. H. S. Ip, and S. Kwong, "Trace back and go ahead: Completing partial annotation for continual semantic segmentation," *Pattern Recognit.*, vol. 165, Sep. 2025, Art. no. 111613.

[8] P. Chen, Y. Ren, B. Zhang, and Y. Zhao, "Class imbalance in the automatic interpretation of remote sensing images: A review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 9483–9508, 2025.

[9] D. Zhou, H. Wang, S. Pang, D. Chen, H. Yao, and J. Yu, "UAV-FAENet: Frequency-aware and attention-enhanced network for remote sensing semantic segmentation of UAV imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 19853–19868, 2025.

[10] L. Qiu, C. Liu, G. Li, X. Dong, F. Xie, and Z. Shi, "Unsupervised deep image stitching for remote sensing: Aligning features across large-scale geometric distortions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 25305–25324, 2025.

[11] M. Pesaresi, A. Gerhardinger, and F. Kayitakire, "A robust built-up area presence index by anisotropic rotation-invariant textural measure," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 1, no. 3, pp. 180–192, Sep. 2008.

[12] H. Wu, S. Yan, C. Liu, A. Wang, M. Wang, and L. Yu, "Hyperspectral open-set classification via frequency-domain rotation enhancement and multibranch adversarial routing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 24864–24882, 2025.

[13] B. Cheng, C. Fan, H. Fu, J. Huang, H. Chen, and X. Luo, "Measuring and computing cognitive statuses of construction workers based on electroencephalogram: A critical review," *IEEE Trans. Computat. Social Syst.*, vol. 9, no. 6, pp. 1644–1659, Dec. 2022.

[14] Y. Wu et al., "ATT-CR: Adaptive triangular transformer for cloud removal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 20595–20610, 2025.

[15] Z. Yu and W. Cui, "Robust hyperspectral image classification using generative adversarial networks," *Inf. Sci.*, vol. 666, May 2024, Art. no. 120452.

[16] L. Huang, Y. Chen, Z. Li, P. Ghamisi, and Q. Du, "HZSCM: Hyperspectral image zero-shot classification via vision-language models," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5528420.

[17] H. Chen, Y. Sun, X. Li, B. Zheng, and T. Chen, "Dual-scale complementary spatial–spectral joint model for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 6772–6789, 2025.

[18] Y. Chu et al., "Global-local graph convolutional broad network for hyperspectral image classification," *Appl. Soft Comput.*, vol. 170, Feb. 2025, Art. no. 112723.

[19] J. Orozco, S. Manzanarez, and V. Manian, "Supervised spatially spectrally coherent local linear embedding—Wasserstein graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5523520.

[20] S. K. Roy, A. Deria, D. Hong, M. Ahmad, A. Plaza, and J. Chanussot, "Hyperspectral and LiDAR data classification using joint CNNs and morphological feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530416.

[21] G. C. Iannelli and P. Gamba, "Urban extent extraction combining sentinel data in the optical and microwave range," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2209–2216, Jul. 2019.

[22] P. Zhong and R. Wang, "Modeling and classifying hyperspectral imagery by CRFs with sparse higher order potentials," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 2, pp. 688–705, Feb. 2011.

[23] M. Ahmad et al., "Spatial–spectral morphological mamba for hyperspectral image classification," *Neurocomputing*, vol. 636, Jul. 2025, Art. no. 129995.

[24] B. Wang, Q. Chen, and Z. Wang, "Diffusion-based visual art creation: A survey and new perspectives," *ACM Comput. Surv.*, vol. 57, no. 10, pp. 1–37, May 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:58:55 UTC from IEEE Xplore. Restrictions apply.

[25] X. Wu, T. Arshad, and B. Peng, "Spectral spatial window attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5519413.

[26] M. Ahmad, M. Mazzara, S. Distefano, A. M. Khan, and S. L. Ullo, "DiffFormer: A differential spatial–spectral transformer for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 10419–10428, 2025.

[27] K. Li and J. C. Príncipe, "Transfer learning in adaptive filters: The nearest instance centroid-estimation kernel least-mean-square algorithm," *IEEE Trans. Signal Process.*, vol. 65, no. 24, pp. 6520–6535, Dec. 2017.

[28] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "A simplified 2D-3D CNN architecture for hyperspectral image classification based on spatial–spectral fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2485–2501, 2020.

[29] S. Minaee, Y. Boykov, F. Porikli, A. Plaza, N. Kehtarnavaz, and D. Terzopoulos, "Image segmentation using deep learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3523–3542, Jul. 2022.

[30] M. E. Paoletti, O. Mogollon-Gutierrez, S. Moreno-Álvarez, J. C. Sancho, and J. M. Haut, "A comprehensive survey of imbalance correction techniques for hyperspectral data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5297–5314, 2023.

[31] Z. Chen, Y. Shen, L. Cao, S. Zhang, and R. Ji, "CLIP-driven transformer for weakly supervised object localization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 6, pp. 4878–4896, Jun. 2025.

[32] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[33] N. A. A. Braham, C. M. Albrecht, J. Mairal, J. Chanussot, Y. Wang, and X. X. Zhu, "SpectralEarth: Training hyperspectral foundation models at scale," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 16780–16797, 2025.

[34] C. Song, Y. Zhang, M. Ma, and S. Mei, "RIFormer+: Rethinking rotation-invariant feature learning in transformer," *IEEE Trans. Multimedia*, vol. 27, pp. 8423–8435, 2025.

[35] L. Song, Z. Feng, S. Yang, X. Zhang, and L. Jiao, "Interactive spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 9, pp. 8589–8601, Sep. 2024.

[36] M. Ahmad, M. Usama, A. M. Khan, S. Distefano, H. A. Altuwaijri, and M. Mazzara, "Spatial–spectral transformer with conditional position encoding for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[37] G. Brauwers and F. Frasincar, "A general survey on attention mechanisms in deep learning," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3279–3298, Apr. 2023.

[38] T. Lin, Y. Wang, X. Liu, and X. Qiu, "A survey of transformers," *AI Open*, vol. 3, pp. 111–132, Nov. 2022.

[39] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[40] H.-C. Tran et al., "Accelerating transformers with spectrum-preserving token merging," in *Proc. Adv. Neural Inf. Process. Syst.* 37, 2024, pp. 30772–30810.

[41] R. Shwartz-Ziv, M. Goldblum, Y. Li, C. B. Bruss, and A. Wilson, "Simplifying neural network training under class imbalance," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 35218–35245.

[42] M. Hahn and M. Rofin, "Why are sensitive functions hard for transformers?" in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics*, 2024, pp. 14973–15008.

[43] R. Azad, M. Dehghanmanshadi, N. Khosravi, J. Cohen-Adad, and D. Merhof, "Addressing missing modality challenges in MRI images: A comprehensive review," *Comput. Vis. Media*, vol. 11, no. 2, pp. 241–268, Apr. 2025.

[44] X. Huang, X. Han, L. Zhang, J. Gong, W. Liao, and J. A. Benediktsson, "Generalized differential morphological profiles for remote sensing image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 4, pp. 1736–1751, Apr. 2016.

[45] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[46] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[47] M. Akewar and M. Chandak, "An integration of natural language and hyperspectral imaging: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 1, pp. 32–54, Mar. 2025.

[48] C. Breining et al., "Acoustic echo control. An application of very-high-order adaptive filters," *IEEE Signal Process. Mag.*, vol. 16, no. 4, pp. 42–69, Jul. 1999.

[49] T. Müller, A. Evans, C. Schied, and A. Keller, "Instant neural graphics primitives with a multiresolution hash encoding," *ACM Trans. Graph.*, vol. 41, no. 4, pp. 1–15, Jul. 2022.

[50] M. Ahmad, H. Luqman, M. Mazzara, and S. L. Ullo, "Saliency-modulated group dynamic attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 1–15, 2026.

[51] A. Jaiswal, T. Chen, J. F. Rousseau, Y. Peng, Y. Ding, and Z. Wang, "Attend who is weak: Pruning-assisted medical image localization under sophisticated and implicit imbalances," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 4976–4985.

[52] A. Jamali, S. K. Roy, D. Hong, P. M. Atkinson, and P. Ghamisi, "Attention graph convolutional network for disjoint hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[53] M. Ahmad and M. Mazzara, "SCSNet: Sharpened cosine similarity-based neural network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–4, 2024.

[54] Y. Wang, L. Yang, T. Xinzhang Wu, K. Tang, and W. Zha, "Spatial–spectral fusion biformer: A novel dynamic routing approach for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5506415.

[55] S. K. Roy, A. Jamali, J. Chanussot, P. Ghamisi, E. Ghaderpour, and H. Shahabi, "SimPoolFormer: A two-stream vision transformer for hyperspectral image classification," *Remote Sens. Appl., Soc. Environ.*, vol. 37, Jan. 2025, Art. no. 101478.

[56] M. Ahmad, U. Ghous, M. Usama, and M. Mazzara, "WaveFormer: Spectral–spatial wavelet transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[57] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "HSI-MFormer: Integrating mamba and transformer experts for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5621916.

[58] Z. Pan, C. Li, A. Plaza, J. Chanussot, and D. Hong, "Hyperspectral image classification with mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5602814.

[59] M. Ahmad, M. Usama, M. Mazzara, and S. Distefano, "WaveMamba: Spatial–spectral wavelet mamba for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[60] A. Yang, M. Li, Y. Ding, L. Fang, Y. Cai, and Y. He, "GraphMamba: An efficient graph structure learning vision mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537414.

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